Rajlaxmi Bhoite

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Google Scholar
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Education

Aalto University — Espoo, Finland
Doctoral Researcher — Feb 2026–Feb 2030

Indian Institute of Science Education and Research — Tirupati, India
BS MS in Physics with minor courses in mathematics (Overall Qualification Good) — 2019–2024

Bharati Vidyapeeth Secondary and Sr.Secondary School — Pune, India
Grade 12 and Grade 10 (Overall Qualification Excellent) — 2008–2019

Experience

SPEC, CNRS, Université Paris-Saclay, CEA Saclay — Paris, France
Research Engineer at Dr. Preden Roulleau's Group — Nov 2024
FQAHE in multilayer graphene **(Link to Agreement)**

RWTH Aachen University — Aachen, Germany
Master's Thesis Student at Prof. Christoph Stampfer Group — August 2023-April 2024
Hybrid Quantum Systems (**Link to MS thesis**)

Tata Institute of Fundamental Research (TIFR) — Mumbai, India
Vigyan Vidushi Student — June 2023-July 2023
Top 40 women students in India 2023 (Physics) out of 500 plus applicants **(Certificate)**

Indian Institute of Science Education and Research (IISER) — Tirupati, India
Project Student at Organic Optoelectronics Lab led by Prof. Kanagasekaran T — January 2023-April 2023
Effect of Dielectric Layer Thickness on Rubrene Based Phototransistors. (**Link**)

Indian Institute of Science (IISC) — Bengaluru, India
Summer Project Intern at Quantum Transport Lab led by Prof. Anindya Das — Summer 2022
Quantum Transport (**Link**)

Qkrishi — New Delhi, India
Summer School Student (Link to Certificate) — Summer 2022
Learning about basics of Quantum Computing · No-Go theorems · Quantum Gates and circuits
Oracle and phase-kickback · Deutsch-Jozsa algorithm · Grover's algorithm **(Link to project)**

Publications

1. Benchmarking Quantum Algorithms on Xanadu, IBM, and Google Quantum Computers
   2022 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT) at Christ University (**Link**)
2. Parametric Amplification in Kerr Nonlinear Resonators: A theoretical review of Josephson Parametric Amplifiers (**Link**)

Projects

- **Summaries on Research Articles** (**Link**)
- **Entanglement Entropy** : Term Paper for Quantum Mechanics 2 (**Link**)
- **Founder Qrivara** : Built a community of young researchers who are passionate about learning and development in fundamental quantum sciences and technologies (**Link**)
- **Automated GUI**: Built an interface to plot graphs and fit them to calculate dephasing and lifetime **(Link)**
- **SQUIDs** : Studying the design and working of DC and RF SQUIDs (**Link**)

Skills

- **Clean Room Fabrication:** Electron Beam Lithography · Reactive Ion Etching · Metal Evaporation · Disco Daddy Dicing Saw · Optical Photolithography · IR Imaging · Raman spectroscopy · Atomic force microscopy · wedge/ball wire bonder · Probe station · Solar simulator · Thermal evaporator · Spin Coater · Glovebox · Parametric Analyser (**Link to Instrument Guide**)
- **Programming Skills :** Python · Fortran · Java · LaTeX· HTML · CSS
- **Softwares :** Gwyddion · Klayout · Corel Draw · VS code · Adobe Illustrator
- **Miscellaneous :** Qiskit Metal **(Link to Qiskit Guide)**· COMSOL & Ansys Q3D · Basics in Cryogenic Measurements · Mathematica · Excel · Origin · Web Development · Data analysis of transport measurements · Scientific writing · Matlab (**Link**) · Machine Learning (**Link**) · Deep Learning (**Link**) · Git

Scholarships/Awards and Invited Talks

- Academic scholarship (Bharati Vidyapeeth, Pune) 2008–2019
- Centor Sector Scholarship during Undergraduate INR 10,000 2019
- Pune Corporation Scholarship INR 25,000 and INR 15,000 2017–2019
- Lab Honararium of INR 20,000 for research internship at IISC 2022
- Funding of INR 7670 from Qkrishi for IEEE conference 2022
- Top 1% National Graduate Physics Examination (NGPE) 2022
- TOEFL iBT - 109/120 2024
- CEA Paris Saclay 5034 Euros June 2025
- Invited talk Institute Neel, Grenoble Quantum Devices and SQC group Jan 2025
- Invited talk FAU Erlangen Prof. Christopher Eichler Group (**Link**) Feb 2025
- InstituteQ Finnish Quantum Days 2026 - Runner up 3MT Competition (500 euros) Sep 2026

Academic References

- Prof. Peter Liljeroth | Aalto University | peter.liljeroth@aalto.fi | PhD supervisor
- Prof.Dr.Anindya Das | IISC India | anindya@iisc.ac.in | MS thesis Advisor
- Prof.Dr.Christoph Stampfer | RWTH Aachen University | stampfer@physik.rwth-aachen.de | MS thesis Advisor
- Dr. Preden Roulleau | CEA Paris Saclay | preden.roulleau@cea.fr | Advisor CEA Saclay